2110575 IoT Systems and Digital Solutions

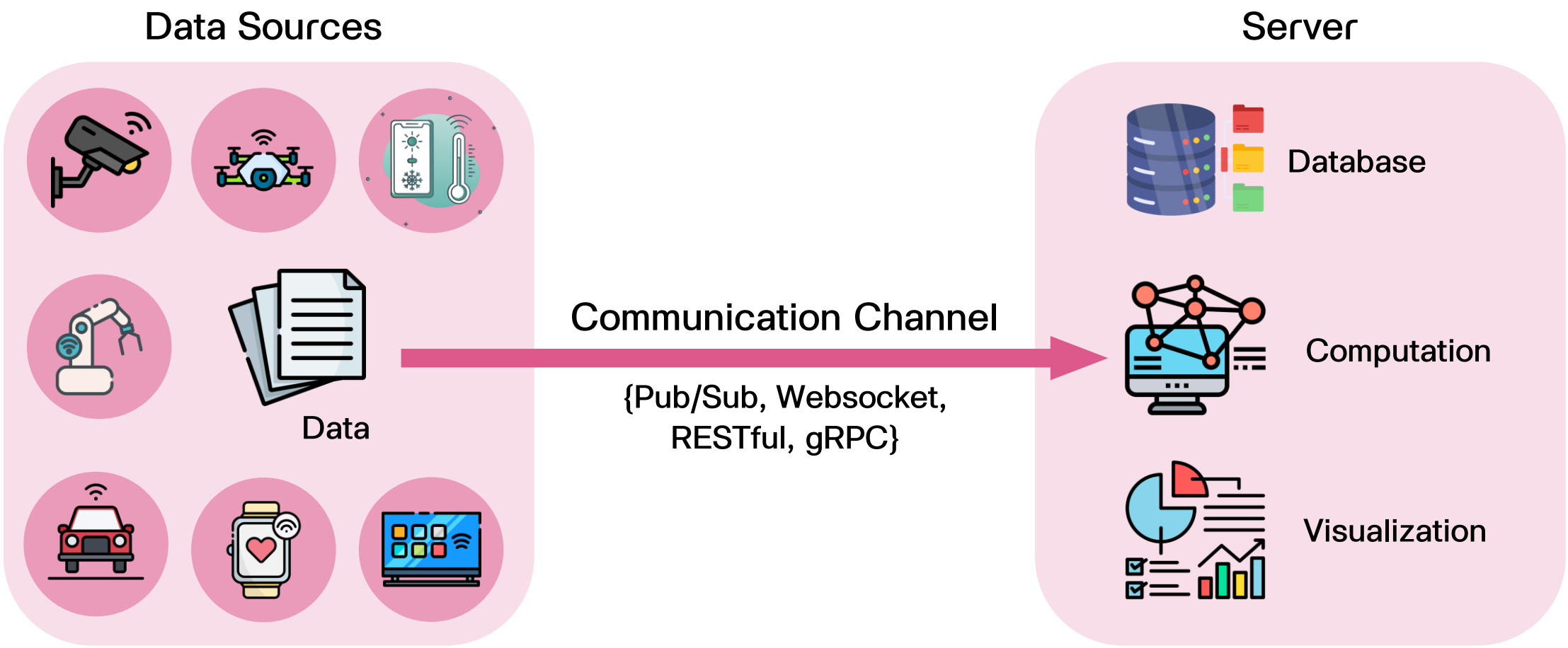
IoT Messaging Platform

Assoc.Prof. Kultida Rojviboonchai

Department of Computer Engineering

Faculty of Engineering, Chulalongkorn University

IOT MESSAGING PLATFORM

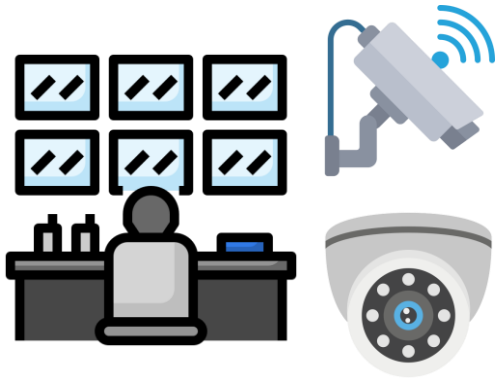


DATA SOURCES

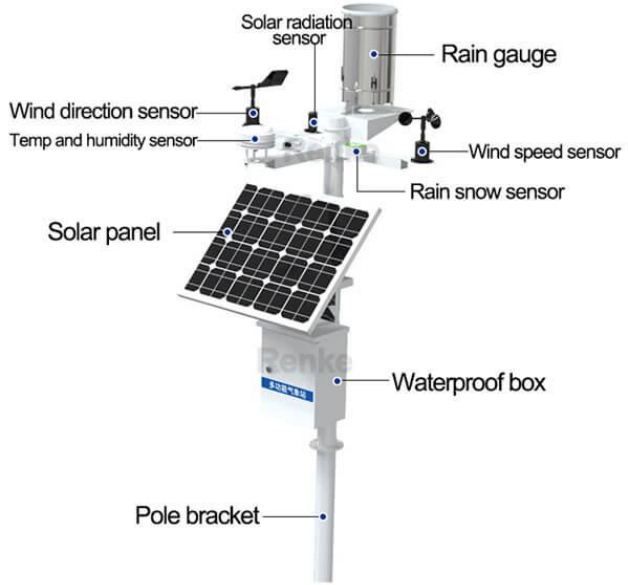
- Generated data from any sources
  - Sensors in vehicles
    - Driving information
  - Sensors in smart city
    - Environmental information
  - Surveillance systems
    - Real-time video











SERVER / DATABASE

- Relational database
  - e.g. MySQL, PostgreSQL, SQLite
- Non-Relational database
  - e.g. Cassandra, Amazon DynamoDB, MongoDB
- Time-Series database
  - e.g. InfluxDB, Amazon Timestream, Prometheus







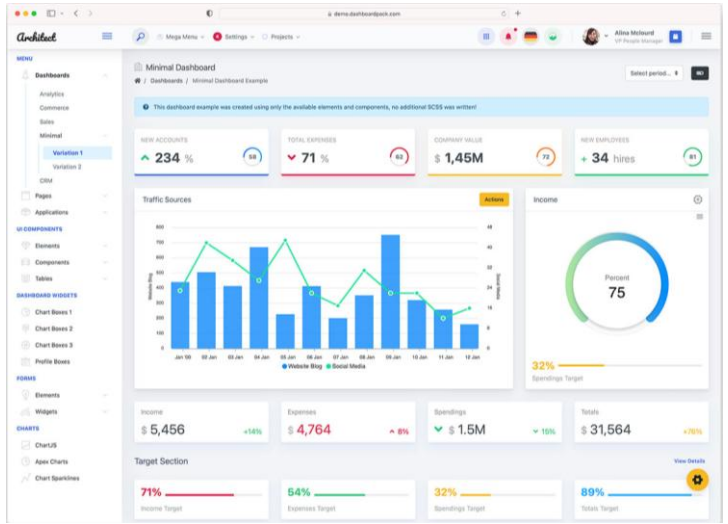


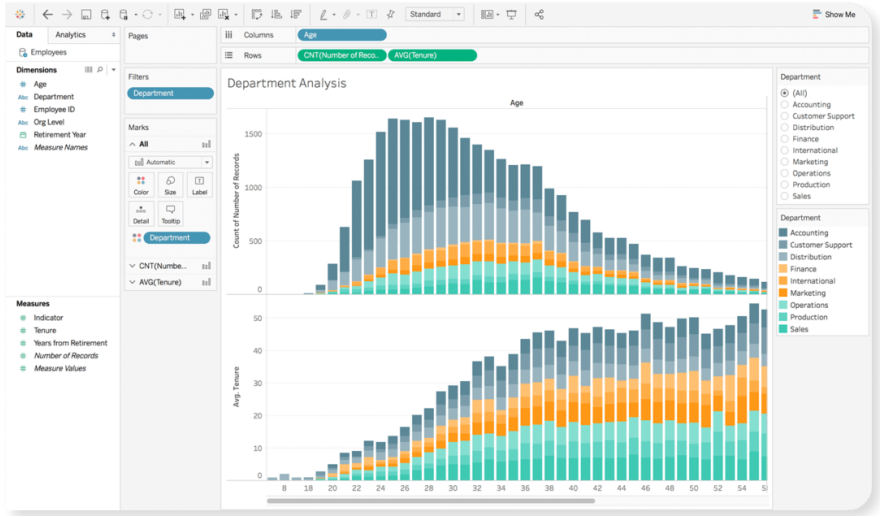


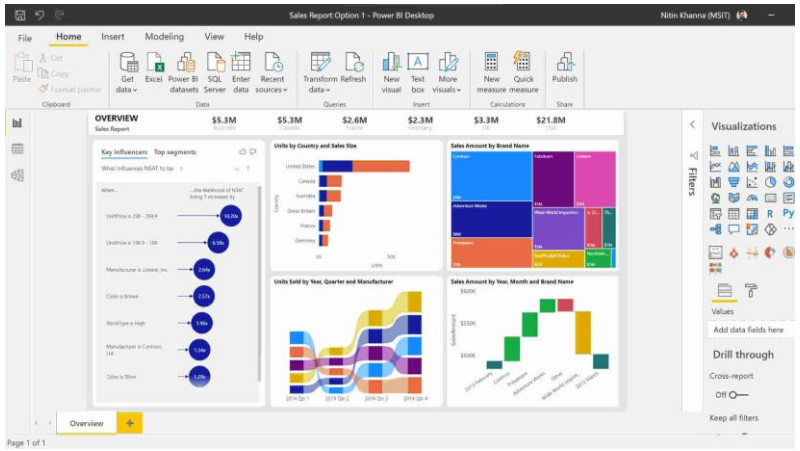


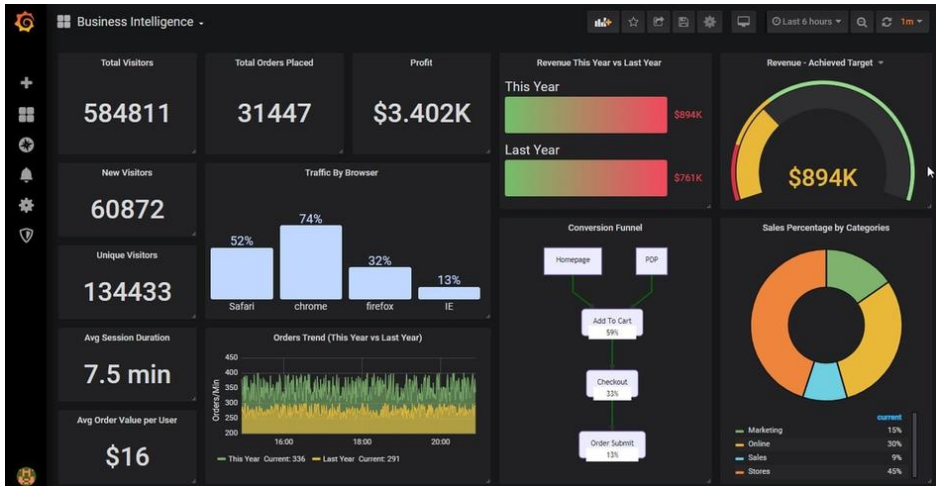
SERVER / VISUALIZATION

- Web application
- Tableau
- QuickSight
- Microsoft Power BI
- Grafana



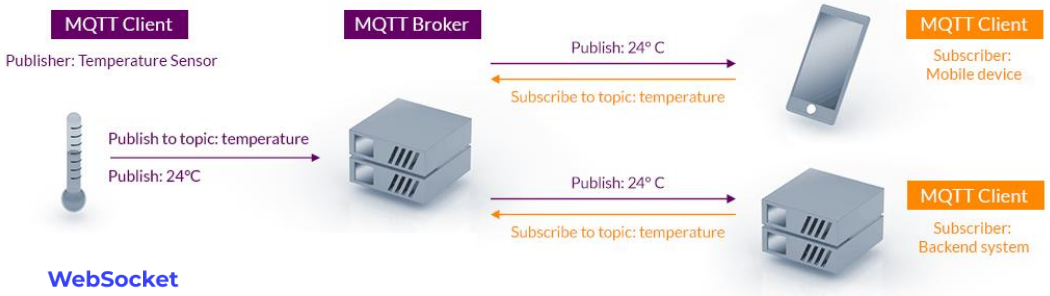


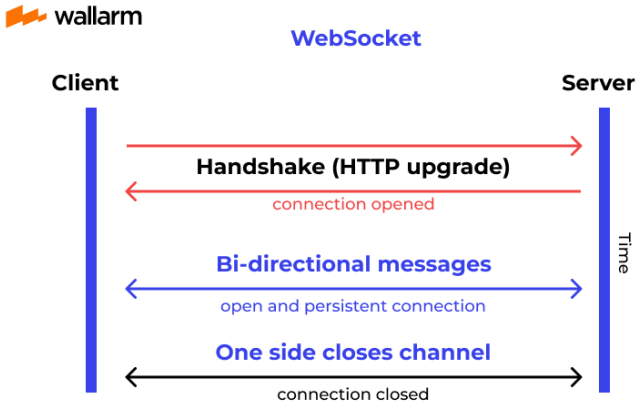


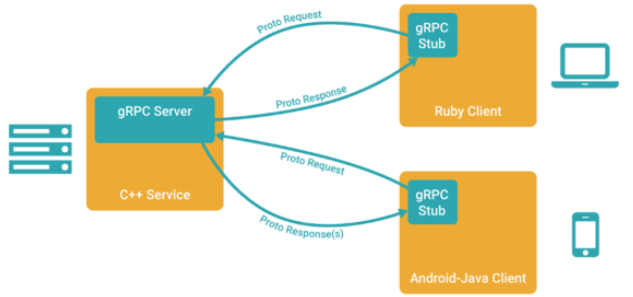


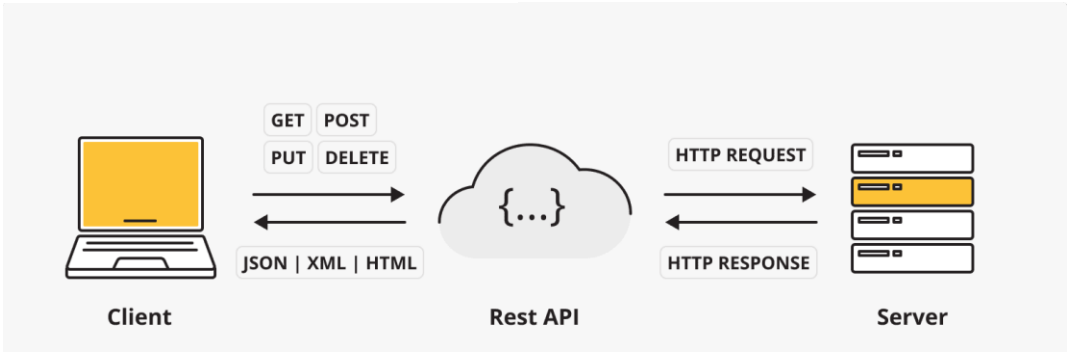
COMMUNICATION CHANNEL

- Pub/Sub
  - MQTT -- https://mqtt.org/
  - Apache Kafka -- https://kafka.apache.org/
- WebSocket
  - https://www.jittagornp.me/blog/what-is-websocket/
- RESTful
  - https://restfulapi.net/
- gRPC
  - https://grpc.io/









PUB/SUB MESSAGING CONCEPT

CONSISTS OF 5 KEYWORDS

- **Message**: the data that is sent through the service.
- **Topic**: a named entity that represents a feed of messages.
- **Publisher** (also called a producer): creates messages and sends (publishes) them to the messaging service on a specified topic.
- **Subscriber** (also called a consumer): receives messages from the messaging service on a specified subscription.
- **Broker**: a central server to which a publisher publishes messages or from which a subscriber consumes messages.
- well-known brokers: Mosquitto, HiveMQ, Apache Kafka, RabbitMQ, Amazon MQ

MQTT – The Standard for IoT Messaging

1. **Lightweight and Efficient**
   - MQTT clients are very small, require minimal resources so can be used on small microcontrollers.
   - MQTT message headers are small to optimize network bandwidth.
2. **Bi-directional Communications**
   - MQTT makes for easy broadcasting messages to groups of things.
3. **Scale to Millions of Things**
   - MQTT can scale to connect with millions of IoT devices
4. **Reliable Message Delivery**
   - MQTT has 3 defined QoS levels: 0 – at most once, 1 – at least once, 2 – exactly once
5. **Support for unreliable Networks**
   - MQTT's support for persistent sessions reduces the time to reconnect the client with the broker
6. **Security Enabled**
   - MQTT makes it easy to encrypt messages using TLS and authentication clients using modern authentication protocols, such as Oauth.

https://mqtt.org/

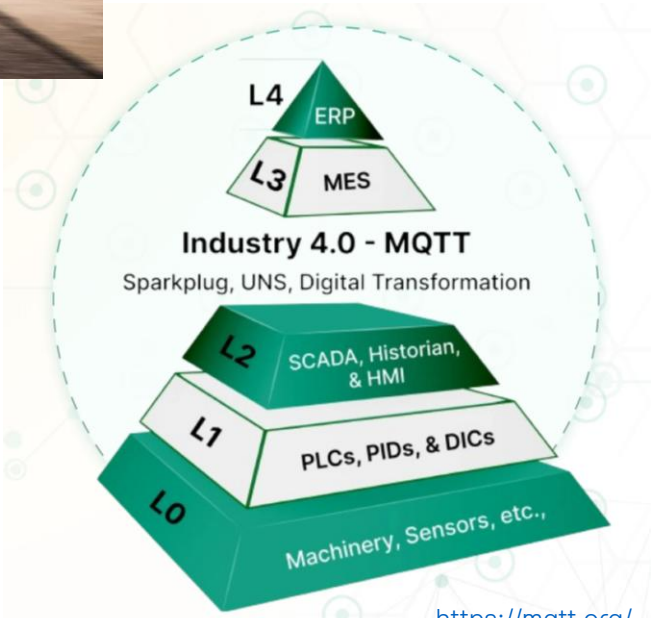
MQTT Use Cases

- **Automotive**
  - HiveMQ: **BMW Car-Sharing application** relies on HiveMQ for **reliable connectivity**
  - EMQ helps **SAIC Volkswagen** building **IoV platform**

Case Study



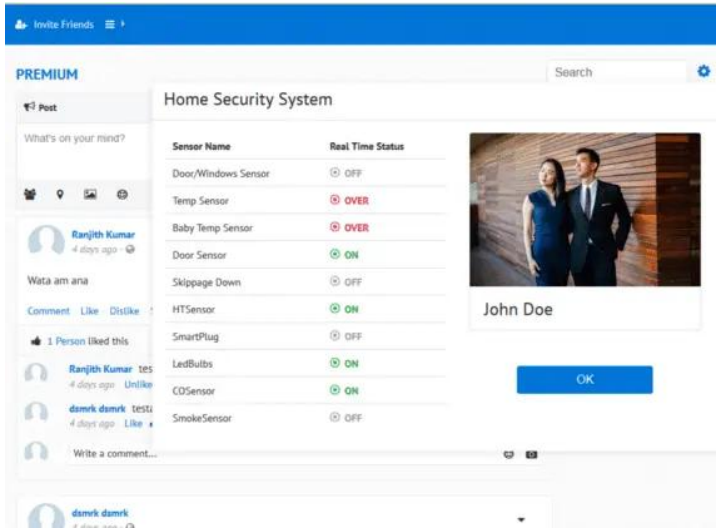
- **Logistics**
  - Transportation & Logistics company **cuts costs and improves asset tracking**
- **Manufacturing**
  - Transforming Manufacturing Efficiency: The Power of MQTT in Industrial Solutions



https://mqtt.org/

MQTT Use Cases

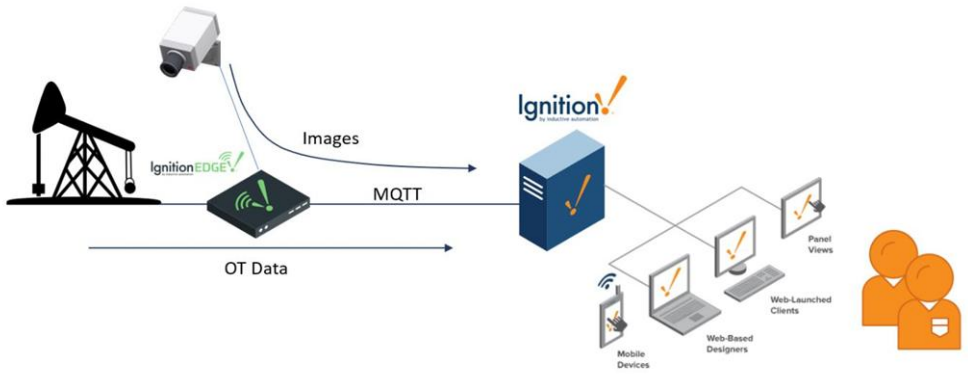
- **Smart Home**
  - The eFon Technology's **Smart Home security system** trusts Bevywise MQTT solution





- **Energy**
  - EMQ helps IoT innovation in the **petrochemical industry**
  - **Energy company** maximizing MQTT for control

- **Consumer Products**
  - CASO Design creates **smart kitchen appliances**



https://mqtt.org/

IoT Messaging Platform Activity